EvoStructCLIP: Multimodal Fusion of Structural and Evolutionary Features for Predicting ATP7B Variant Effects

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I. Data sources

To improve generalization, we performed training on a large pathogenicity dataset curated from ClinVar [2]. We downloaded all human variants and filtered for missense variants with unambiguous pathogenic or benign labels. Quality control included mapping to canonical UniProtKB isoforms, sequence consistency checks using Swiss-Prot and TrEMBL reference FASTA files [3], structure mapping to AlphaFold DB models [4], and removal of duplicates. This yielded 153,787 unique missense variants with pathogenicity labels.

Multiple sequence alignments (MSAs) of homologous sequences were generated using MMseqs2 [5], and evolutionary descriptors (e.g., conservation, entropy) were derived from these MSAs. Structural information was obtained from AlphaFold DB release v4 [4].

II. Feature extraction

A. Voxel-based structural features

We generated three-dimensional voxel representations centered on the Cα atom of the mutated residue. A cubic grid of size $7 \times 7 \times 7$ with a spacing of 2 Å was defined, and each voxel was annotated with multiple structural descriptors. First, 42 "closeness" channels captured the proximity of CA and CB atoms from each of the 21 standard amino acids, following the general strategy of PrimateAI-3D [6] but extended with additional features. Relative sequence position was encoded by normalized signed and absolute distances between the mutated residue and the residue mapped to each voxel. Per-residue AlphaFold confidence scores (pLDDT) were projected onto the grid to reflect model reliability. Finally, local dynamic flexibility was quantified by mean square fluctuations derived from Gaussian Network Model (GNM) analysis using ProDy [8]. Together, these features produced approximately 46 channels per voxel grid, offering a richer structural representation than the original voxelization approach.

B. Evolutionary features

Evolutionary constraints were captured using multiple sequence alignments (MSAs) of homologous proteins generated with MMseqs2 [5]. MSAs were filtered by sequence identity, coverage, and redundancy thresholds to ensure high-quality alignments. From these MSAs, we derived conservation scores, amino acid substitution preferences, and position-specific information content, which provided complementary evolutionary context to the structural representations.

III. Model architecture (EvoStructCLIP, pretraining)

A. Overall framework

We developed EvoStructCLIP, a multimodal architecture designed to integrate structural and evolutionary evidence for missense variant effect prediction. The model contains two branches: a voxel-based encoder that captures local three-dimensional structural context around the mutated residue, and an MSA-based encoder that leverages evolutionary variation across homologous sequences. These branches produce embeddings that are aligned through a CLIP-style contrastive objective, while a classification head predicts pathogenicity [13].

B. Voxel branch

The voxel encoder processes the 3D voxelized structural features described earlier. The backbone is constructed from 3D MBConv blocks, which combine pointwise expansion, depthwise convolution, and squeeze-and-excitation attention [19,20]. This design allows the network to capture both fine-grained local interactions and broader spatial contexts while maintaining computational efficiency, consistent with principles from EfficientNet [11]. After global average pooling, the output is enriched with mutation-specific embeddings: both the wild-type and mutant residues are embedded, concatenated, and passed through non-linear transformations. These embeddings are added to the pooled structural representation, ensuring that the final vector explicitly encodes both the local structural context and the residue substitution itself.

C. MSA branch with cross-axial Mamba

The MSA encoder models evolutionary information from sequence alignments centered on the mutated position. Each amino acid is first embedded into a continuous space, and the resulting tensor has shape (L, D, C), where L is the alignment length window, D the MSA depth, and C the embedding dimension. To capture dependencies, we introduced a cross-axial Mamba block. Along the sequence length (L-axis), a full Mamba state-space layer processes each column, enabling long-range context propagation that is more efficient than classical recurrent or attention-based encoders [12,22]. Along the alignment depth (D-axis), convolutional filters are applied only at the central sequence position corresponding to the mutated residue, extracting local consensus signals across homologs. By combining outputs from both axes, the encoder integrates positional conservation with depth-wise evolutionary diversity in a complementary fashion [23]. Residual connections and RMS normalization stabilize training [24], and the final embedding is obtained by averaging the depth dimension at the mutation site.

D. Fusion and contrastive alignment

The voxel and MSA embeddings are L2-normalized and compared in a similarity matrix within each batch. A temperature-scaled CLIP objective encourages embeddings from the same variant to align closely while mismatched pairs are pushed apart, enforcing cross-modal consistency [13]. For downstream classification, the embeddings are either concatenated or added, followed by a lightweight feed-forward network with batch normalization and non-linear activation, outputting a scalar prediction score.

E. Training strategy

The model was trained end-to-end with a composite loss consisting of binary cross-entropy for pathogenicity, the CLIP contrastive loss [13], and an auxiliary FuseMix loss. FuseMix interpolates pairs of voxel and MSA embeddings in latent space using a shared Beta-distributed coefficient across modalities, ensuring that the augmented samples remain semantically consistent multimodal pairs. This latent-space augmentation alleviates data scarcity and improves robustness [14,25]. Training employed the AdamW optimizer [26] with cosine annealing learning rate scheduling [27] over 100 epochs. Early stopping was guided by validation performance using precision–recall AUC [28]. Data augmentations included random voxel rotations and flips [29], as well as MSA depth shuffling to prevent overfitting to alignment ordering.

IV. DOWNSTREAM STABILITY PREDICTION

To adapt EvoStructCLIP for quantitative prediction of ATP7B variant fitness in the yeast complementation assay, we directly utilized the sigmoid-activated pathogenicity scores produced by the pretrained EvoStructCLIP model. For each variant, the predicted probability of pathogenicity was transformed into a fitness score by taking its complement (fitness = 1 − pathogenicity score), such that 0 represents no growth, 1 indicates wild-type-like growth, and values greater than 1 would correspond to improved fitness.

Because our model was designed for binary classification of pathogenic versus benign variants, no additional regression or handcrafted feature integration was performed. Standard deviations were set to zero, reflecting that only a single model was used without ensemble uncertainty estimation.

This straightforward adaptation allowed EvoStructCLIP to generate continuous fitness predictions across all ATP7B variants in the challenge dataset, providing a simple baseline submission consistent with the evaluation framework of the CAGI7 ATP7B challenge. Nonsense (Ter) variants were assigned 0.0 fitness heuristically, while synonymous variants were left as "*" according to submission guidelines.

In addition to the baseline submission (UXFactory_model_1), we explored several post-hoc scaling strategies to adjust the distribution of predicted scores. Specifically, UXFactory_model_2 applied a square-root transformation ($\alpha = 0.5$), UXFactory_model_3 used a milder scaling with $\alpha = 0.7$, and UXFactory_model_4 employed a more aggressive scaling with $\alpha = 0.3$. These variants were included to evaluate whether simple monotonic transformations of the baseline predictions could improve correlation with experimental fitness values.

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